

(Ubuntu Wiki, 2019). Usually, *mq-deadline* and *none* are available by default. If the other I/O schedulers are missing you could use the following commands:

```
$ sudo modprobe bfq
$ sudo modprobe kyber-iosched
```

Now run the following command to check the available I/O schedulers:

```
$ cat /sys/block/sda/queue/scheduler
```

The output would be something like:

```
[mq-deadline] none bfq kyber
```

Let's now move on to the benchmarks.

Test 1: writes-starving-reads

This test evaluates the performance of the I/O schedulers when there is a write operation going on in the background and how efficiently the read operations can be completed without starvation. Benchmark Test 1 is designed in a way to get results for better comparison.

You can perform the write operation in the background, using the following command:

```
$ while true; do dd if=/dev/zero of=file bs=1M; done &
```

This command runs a write operation in the background due to the '&' symbol used at the end of the command. The above command is an infinite *while* loop.

After the testing on benchmarks is done, remember to kill this process using its process ID.

The code for changing to the *none* scheduler and then running Benchmark Test 1 is given below.

Step 1: Perform the write operation in the background.

Step 2: Find out how long does it

Table 1: *none* I/O scheduler results

File size	Real-time in Ubuntu 20.04 (in seconds)	Real-time in Docker and container (in seconds)	Percentage improvement
10MB	0.003	0.002	33.33
20MB	0.005	0.003	40.00
30MB	0.006	0.005	16.67
40MB	0.008	0.006	25.00
50MB	0.009	0.007	22.22
60MB	0.01	0.008	20.00
70MB	0.012	0.01	16.67
80MB	0.013	0.011	15.38
90MB	0.015	0.012	20.00
100MB	0.017	0.013	23.53

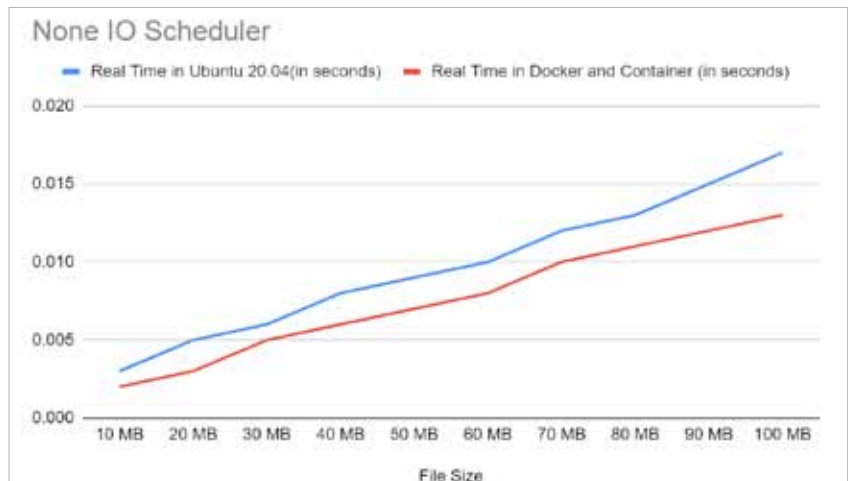


Figure 1: Graphical performance comparison of *none* I/O schedulers in VM vs Docker container

Table 2: *mq-deadline* I/O scheduler results

File size	Real-time in Ubuntu 20.04 (in seconds)	Real-time in Docker and container (in seconds)	Percentage improvement
10MB	0.012	0.002	83.33
20MB	0.018	0.003	83.33
30MB	0.024	0.004	83.33
40MB	0.036	0.005	86.11
50MB	0.046	0.007	84.78
60MB	0.051	0.008	84.31
70MB	0.059	0.01	83.05
80MB	0.062	0.012	80.65
90MB	0.072	0.013	81.94
100MB	0.076	0.013	82.89